CPU Scheduling Simulator

Operating Systems Lab Assignment

NAME: NITU

REG.NO. :24MIP10006

1. Introduction

CPU Scheduling is an important concept in operating systems. It determines how the CPU selects a process from the ready queue for execution. Efficient scheduling improves system performance, reduces waiting time, and increases CPU utilization.

This project implements a CPU Scheduling Simulator that demonstrates different scheduling algorithms and compares their performance.

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2. Objectives

- To understand CPU scheduling algorithms
- To implement different scheduling techniques
- To calculate Waiting Time and Turnaround Time
- To visualize execution using Gantt Chart
- To compare performance of different algorithms

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3. Algorithms Implemented

3.1 First Come First Serve (FCFS)

- Processes are executed in the order of arrival
- Simple and easy to implement
- May cause long waiting time (convoy effect)

3.2 Shortest Job First (SJF) – Non-Preemptive

- Process with smallest burst time is executed first
- Reduces average waiting time
- Can cause starvation for longer processes

3.3 Round Robin (RR)

- Each process is given a fixed time quantum
- Ensures fairness among processes
- Suitable for time-sharing systems

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4. Performance Metrics

Turnaround Time (TAT)

Turnaround Time = Completion Time – Arrival Time

Waiting Time (WT)

Waiting Time = Turnaround Time – Burst Time

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5. Output Results

- **FCFS Output**

```
------------------------
1. FCFS
2. SJF
3. Round Robin
4. SRTF
Enter your choice: 1
Enter number of processes: 4
Enter Arrival Time for P1: 3
Enter Burst Time for P1: 3
Enter Arrival Time for P2: 44
Enter Burst Time for P2: 3
Enter Arrival Time for P3: 2
Enter Burst Time for P3: 5
Enter Arrival Time for P4: 6
Enter Burst Time for P4: 7

Running FCFS...


Gantt Chart:
| P3 | P1 | P4 | P2 |
0   7   10   17   47


PID     AT      BT      CT      TAT     WT
P3      2       5       7       5       0
P1      3       3       10      7       4
P4      6       7       17      11      4
P2      44      3       47      3       0

Average Turnaround Time: 6.5
Average Waiting Time: 2.0
PS N:\SEM 5 WINTER\OPERATING SYSTEM\CPU_SCHEDULING_PROJECT>
```

- **SJF Output**

```
CPU Scheduling Simulator
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1. FCFS
2. SJF
3. Round Robin
4. SRTF
Enter your choice: 2
Enter number of processes: 4
Enter Arrival Time for P1: 6
Enter Burst Time for P1: 7
Enter Arrival Time for P2: 5
Enter Burst Time for P2: 8
Enter Arrival Time for P3: 3
Enter Burst Time for P3: 4
Enter Arrival Time for P4: 3
Enter Burst Time for P4: 2

Running SJF...


Gantt Chart:
| P4 | P3 | P1 | P2 |
0   5   9   16   24


PID     AT      BT      CT      TAT     WT
P1      6       7       16      10      3
P2      5       8       24      19      11
P3      3       4       9       6       2
P4      3       2       5       2       0

Average Turnaround Time: 9.25
Average Waiting Time: 4.0
PS N:\SEM 5 WINTER\OPERATING SYSTEM\CPU_SCHEDULING_PROJECT>
```

- **Round Robin Output**

```
PS N:\SEM 5 WINTER\OPERATING SYSTEM\CPU_SCHEDULING_PROJECT> python -u "n:\SEM 5 WINTER\OPERATING SYSTEM\CPU_SCHEDULING_PROJECT\main.py"
1. FCFS
2. SJF
3. Round Robin
4. SRTF
Enter your choice: 3
Enter number of processes: 4
Enter Arrival Time for P1: 5
Enter Burst Time for P1: 3
Enter Arrival Time for P2: 7
Enter Burst Time for P2: 4
Enter Arrival Time for P3: 6
Enter Burst Time for P3: 7
Enter Arrival Time for P4: 4
Enter Burst Time for P4: 7
Enter Time Quantum: 4

Running Round Robin...


Gantt Chart:
| P4 | P1 | P3 | P2 | P4 | P3 |
0   8   11   15   19   22   25


PID     AT      BT      CT      TAT     WT
P4      4       7       22      18      11
P1      5       3       11      6       3
P3      6       7       25      19      12
P2      7       4       19      12      8

Average Turnaround Time: 13.75
Average Waiting Time: 8.5
PS N:\SEM 5 WINTER\OPERATING SYSTEM\CPU_SCHEDULING_PROJECT>
```

- **SRTF OUTPUT**

```
3. Round Robin
4. SRTF
Enter your choice: 4
Enter number of processes: 6
Enter Arrival Time for P1: 3
Enter Burst Time for P1: 4
Enter Arrival Time for P2: 7
Enter Burst Time for P2: 3
Enter Arrival Time for P3: 4
Enter Burst Time for P3: 6
Enter Arrival Time for P4: 3
Enter Burst Time for P4: 9
Enter Arrival Time for P5: 8
Enter Burst Time for P5: 5
Enter Arrival Time for P6: 6
Enter Burst Time for P6: 3

Running SRTF...


Gantt Chart:
| P1 | P2 | P6 | P5 | P3 | P4 |

PID     AT      BT      CT      TAT     WT
P1      3       4       7       4       0
P2      7       3       10      3       0
P3      4       6       24      20      14
P4      3       9       33      30      21
P5      8       5       18      10      5
P6      6       3       13      7       4

Average Turnaround Time: 12.333333333333334
Average Waiting Time: 7.333333333333333
PS N:\SEM 5 WINTER\OPERATING SYSTEM\CPU_SCHEDULING_PROJECT>
```

Each output shows:

- Completion Time
- Turnaround Time
- Waiting Time
- Gantt Chart

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6. Challenging Task

6.1 Shortest Remaining Time First (SRTF)

- Pre-emptive version of SJF
- Process with smallest remaining burst time is selected
- If a shorter process arrives, current process is interrupted

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6.2 Comparison of Algorithms

Algorithm		Avg Waiting Time	Avg Turnaround Time
FCFS		2.0	6.5
SJF		4	9.25
RR		8.5	13.75

SRTF		7.3	12.33

7. Conclusion

This project helped in understanding how different CPU scheduling algorithms work.

- FCFS is simple but not efficient
- SJF provides better performance but may cause starvation
- Round Robin ensures fairness
- SRTF gives optimal waiting time but is complex

Overall, SJF and SRTF provide better performance in terms of waiting and turnaround time.

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8. How to Run the Program

1. Open terminal
2. Run the command:
   python main.py
3. Enter process details as required
4. Select the scheduling algorithm
5. View output and Gantt Chart

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9. Tools Used

- Python
- VS Code